# Multimodal AI Teaching Copilot: Live Slide Generation from Lecture Speech

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***Abstract*—Prepared slides cannot show what a teacher adds while explaining, and a large language model (LLM) that regenerates slides from a growing transcript rewrites what students are reading. This paper presents a teaching copilot that builds a structured, continuity-aware slide deck on the classroom projector from the teacher's live speech. Locally transcribed speech is sent in bounded units to an LLM that only returns a schema-validated interpretation; a deterministic planner turns it into slide operations that only extend the deck, so the LLM cannot rewrite what students have read, and a transcript-coverage guard restores facts the LLM omits. Across 36 held-out lectures (409 min), run on one laptop with 4 GB of graphics memory, the median delay from speech to slide was 11.2 s. On an annotated subset of 18 lectures, slide changes matched topic boundaries with precision 0.78, 86.6 % of key facts reached a slide, 0.9 % of statements were unsupported by the speech, and five teachers gave mean ratings of 3.87–4.17 of 5 on four criteria. On the same units, an LLM-controlled baseline rewrote or removed 38.2 % of the statements students had already seen, against 2.0 % (lower in all 18 lectures, $p < 0.001$), and a replay ablation showed that opening a slide per concept lowers precision to 0.37.**

***Index Terms*—Automatic slide generation, educational technology, large language models, live systems, speech recognition, human-in-the-loop**

## I. Introduction

Slides prepared before a lecture fix its order in advance, so the examples and answers a teacher adds while explaining never reach the screen. Narration paired with relevant pictures helps learners more than narration by itself [1], provided the visuals add no extraneous load [2], [3]; yet no teacher can draw such visuals while teaching.

Existing tools solve only parts of this problem: live captions show verbatim text without the structure of the lesson [4]; slide generators work offline from a finished document or prompt [5]–[11]; and a large language model (LLM) [12], [13] applied to a growing transcript rewrites slides while students read them and adds untaught content [14]. Three problems remain open: (P1) projected material does not follow what is taught; (P2) live speech is not turned into a structured visual lesson; and (P3) LLM output is neither stable nor faithful to the teacher's words.

This paper presents the Multimodal AI Teaching Copilot, an implemented system that addresses all three: spoken input becomes visual output (typed layouts, typeset formulas, diagrams and retrieved images) on the projector while the teacher speaks. The contributions are: (1) incremental slide generation from live speech; (2) an extend-only update policy: the LLM only labels bounded, schema-validated units, and a deterministic planner executes only operations that add to the deck, so the LLM cannot rewrite what students have read; (3) a transcript-coverage guard that restores facts the LLM omits, and a formula guard; and (4) a held-out evaluation with annotation, teacher ratings, a replay ablation and a paired comparison with an LLM-controlled baseline.

## II. Related Work

### A. Multimedia Learning and Lecture Speech

In Mayer's experiments, learners given text with matching pictures outperformed those given text only [1]; Sweller attributes such effects to how little a learner can hold in mind at once [2], [3], and decorative pictures reduced test scores in PowerPoint lectures [15]. The system therefore uses short typed layouts and adds an image only for a concrete object that is taught, vetted with CLIP, which Radford *et al.* trained on image–caption pairs so that any picture can be scored against a text prompt [16].

Whisper by Radford *et al.* [17], run with 8-bit weights by faster-whisper [18], and Silero VAD [19] form the speech front end. Kushalnagar *et al.* showed that verbatim real-time captions are hard for students to follow [4], and Gonzalez *et al.* found that LLM summaries of recorded lectures improved quiz scores [20]; both produce text, not a structured visual lesson (P2).

### B. Segmentation, Slide Generation and LLMs

TextTiling by Hearst places subtopic boundaries where the similarity of adjacent blocks drops [21]; Sentence-BERT by Reimers and Gurevych compares sentence embeddings by cosine similarity [22], and Wang *et al.* distilled it into MiniLM [23]. These methods segment a complete transcript offline; here the similarity drop is one online signal. Boundaries are scored with the $P_k$ measure of Beeferman *et al.* [24], and fact coverage with the word-recall overlap that ROUGE-1 by Lin uses for summaries [25].

PPSGen selects sentences of a paper for slides by integer linear programming [7], D2S summarizes document text for each slide title [6], and DOC2PPT generates slides with text and figures from scientific documents [5]. With LLMs, Xie *et al.* ground educational slides in retrieved sources [8], SlideBot adds multi-agent planning [9], PPTAgent edits reference slides [10], and AutoPresent writes slides as code [11]. All build a whole deck offline from written input; a lecture arrives incrementally as noisy speech (P1, P3).

TABLE I
COMPARISON OF THE PROPOSED SYSTEM WITH PRIOR SYSTEMS

| System | Input | Produced | Topic structure | Faithfulness to source | Slide sequence |
|---|---|---|---|---|---|
| Live captions [4], [17] | Speech | Live | None (text stream) | Verbatim text | None |
| Lecture summaries [20] | Recorded lecture | Offline | None (summary) | LLM summary | None |
| PPSGen, D2S, DOC2PPT [5]–[7] | Paper / document | Offline | Document sections | Extracted from document | Whole deck at once |
| LLM slide generators [8]–[11] | Topic, document | Offline | LLM outline / plan | Retrieved sources or none | Whole deck at once |
| **Proposed system** | **Live speech** | **During lecture** | **Online topic / subtopic** | **Coverage + formula guards** | **Extend-only + teacher** |

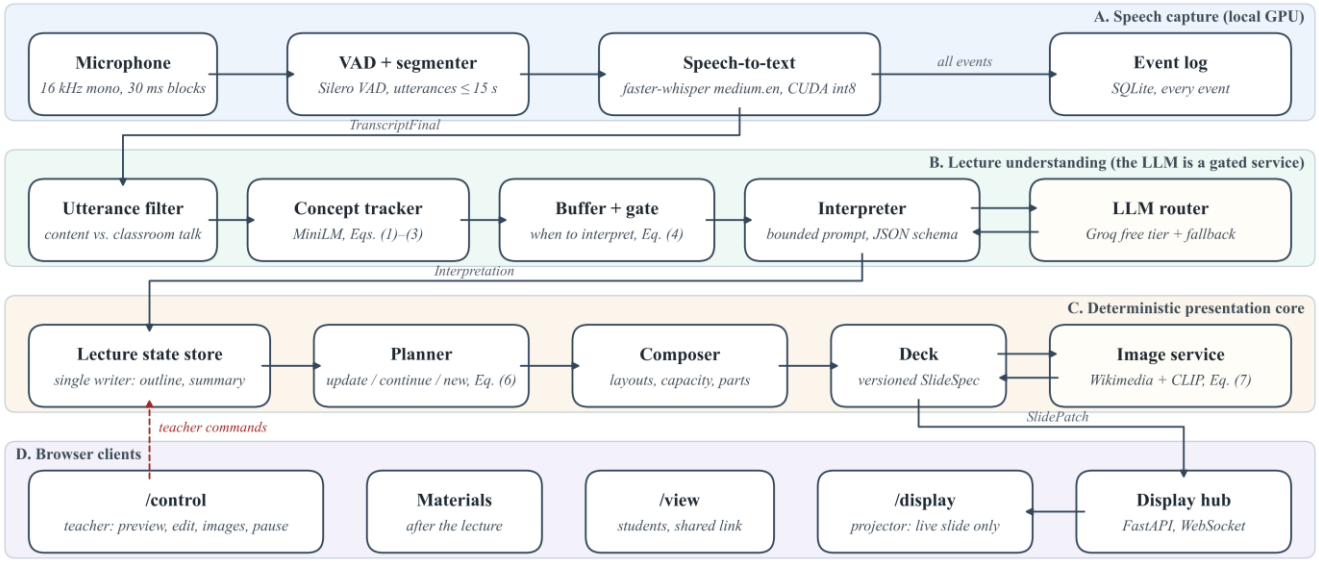



Fig. 1. Implemented architecture. Lanes A–C run in one Python process; only the interpreter calls the LLM; browsers get slide patches and send commands.

Kasneci *et al.* stressed teacher oversight of LLMs in education [12], so the teacher overrides every automatic decision. Ji *et al.* surveyed hallucination [14], and Willard and Louf constrained generation to a JSON schema [26], which fixes the format but not the facts, so the output is also checked against the transcript. Open-weight models, gpt-oss-120b [13] and the Qwen3 family [27], serve as interpreters on a free hosted tier; generated quizzes follow the review of Kurdi *et al.* [28]. Annotator agreement uses Cohen's κ [29] and Gwet's AC1, which stays informative when one label dominates [30]; rater consistency uses the ICC(2,1) of Shrout and Fleiss [31].

Among the systems in Table I, none combines streaming speech input with incremental decisions about which slide to change while students read; this is the gap addressed here.

## III. METHODOLOGY

### A. System Overview

A lecture is processed as a stream through six stages (Fig. 1): speech capture and recognition; filtering and concept tracking; a gate that groups finished sentences into units; bounded LLM interpretation of each unit; a deterministic planner and composer that turn it into slide operations and layouts; and delivery to the projector, the teacher and students. Four rules guide the design: the LLM supplies labels but never executes a slide operation, it never sees the whole transcript, a new concept is not by itself a new slide, and user interfaces send commands instead of changing the state.

All components run in one Python *asyncio* process and exchange typed events; every event is logged in SQLite, from which each lecture is replayed for evaluation and materials. Recognition runs on the GPU, MiniLM and CLIP on the CPU (ONNX), and the LLM on the Groq free tier (openai/gpt-oss-120b [13]; qwen/qwen3.8-27b of the Qwen3 family [27] when that quota is spent; Oct. 2026), as a 7B-class local model does not fit beside the speech model in 4 GB. Of the 1,619 held-out units, 71.6 % were interpreted by the first model, 24.0 % by the second and 4.4 % by the deterministic fallback.

### B. Speech Capture and Concept Tracking

Silero VAD [19] segments 16 kHz audio (utterances up to 15 s), and faster-whisper [18] with Whisper medium.en [17] (int8, beam size 1, keyword prompt) transcribes them. A deterministic filter removes classroom management, meta talk and fillers. Each content utterance is embedded with MiniLM [23]; with the L2-normalized embedding $e_t$ and concept centroid $c_{t-1}$, the shift score is the cosine distance $s_t = 1 - e_t \cdot c_{t-1}$ used with Sentence-BERT [22], and the centroid is the exponentially weighted mean $c_t = \mathrm{norm}((1-\alpha)c_{t-1} + \alpha e_t)$ with $\alpha = 0.35$, raised to 0.6 at a boundary so it re-seeds on the new concept. Following the similarity-drop criterion of TextTiling [21], a boundary is signalled when

$$b_t = \kappa_t \vee (s_t \geq \tau), \tau = 0.75 \quad (1)$$

where $\kappa_t$ marks a cue phrase ("moving on to", "next"); $\tau$ is high because shift scores within one facet were 0.3–0.6.

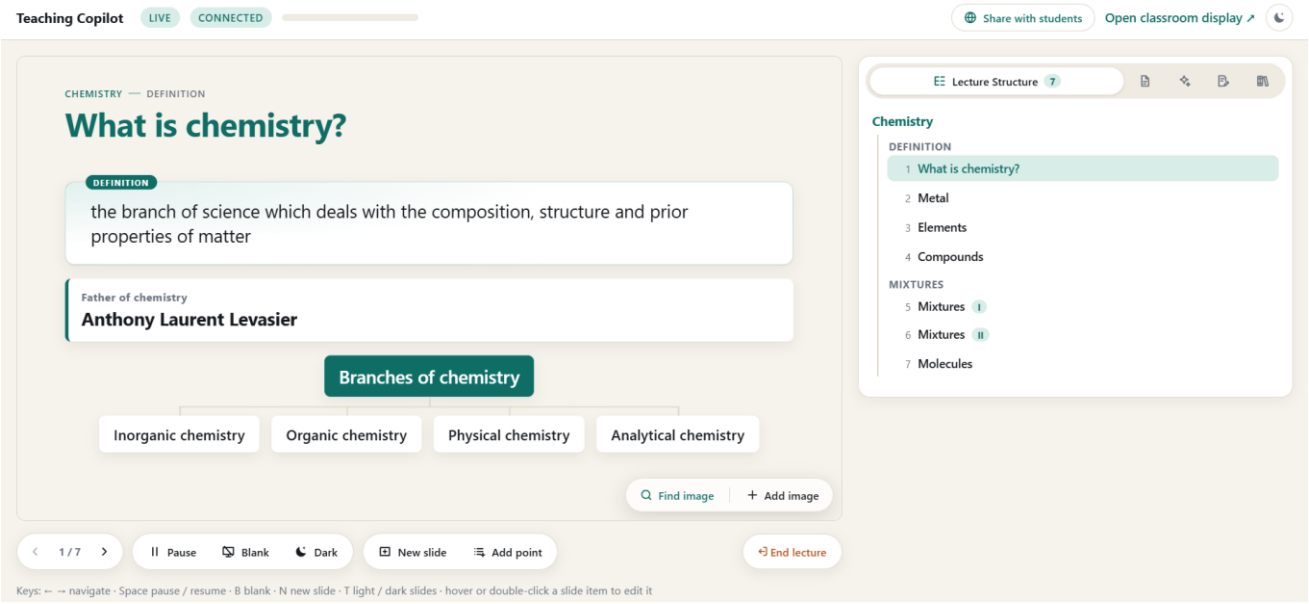



Fig. 2. Teacher control view (/control) in a live chemistry lecture: the projected slide as an editable preview with image tools and lecture controls (left) and the lecture structure tree built by the system, topic → facet → slides, with overflow parts I and II (right).

### C. Gate and Token Budget

Content accumulates in a buffer until a rule-based gate fires. With $p_t$ the silence after a finished sentence, $w_t$ the buffered words and $a_t$ the age of the oldest pending line (s),

$$G_t = (p_t \geq 1.2 \wedge w_t \geq 6) \vee w_t \geq 40 \vee b_t \vee a_t \geq 12 \quad (2)$$

with an 8 s floor between calls, set by the free quota (200,000 tokens per key and model per day).

### D. Bounded Interpretation and Transcript-Grounding Safeguards

The prompt holds the subject, level, an outline (≤ 150 tokens), the current slide with numbered items, a rolling summary (≤ 120 tokens) and the new lines, within 3,000 tokens (temperature 0.2). The LLM returns JSON with topic, subtopic, a relation (same concept, elaboration, sub-concept, sibling, new topic, digression), typed content, an image hint and *concerns* about possibly wrong statements. Pydantic validates it; after one failed repair, a deterministic fallback shows the spoken sentences as points. A coverage guard compares every content sentence $u$ (≥ 6 words) with the output $O$; with $W(\cdot)$ the set of word stems and a recall-style overlap as in ROUGE-1 [25],

$$|W(u) \cap W(O)| \leq \tfrac{1}{3}|W(u)| \Rightarrow u \text{ is shown as said} \quad (3)$$

so a fact the model left out still reaches the slide. A formula guard flags changed formula tokens, and a correction is applied only above a confidence of 0.75 (factual) or 0.4 (mis-hearing); otherwise the slide shows what was said and the teacher gets a card with both versions. These checks keep slides faithful to the transcript; they do not prove a statement correct.

### E. Slide Planner

The planner turns each interpretation into one operation: *update* the working slide, *continue* on a new slide of the same topic, open a *new* topic slide, *retitle*, or nothing. None of them lets the LLM rewrite or delete a shown item: shown text changes only when the composer merges an elaboration into the item it elaborates, a mis-hearing correction is applied, or the teacher edits it. A topic change must be confirmed, because a single unit may be misread:

$$C_n = B_n \vee \left(\bar{s}_n^{\max} \geq \tau\right) \vee \left(T_n \equiv \tilde{T}_{n-1}\right) \quad (4)$$

where $B_n$ is true if a boundary (1) occurred in unit $n$, $\bar{s}_n^{\max}$ is its largest shift score smoothed over two utterances, $T_n$ its topic and $\tilde{T}_{n-1}$ an unconfirmed candidate from the previous unit. Unconfirmed content is shown tentatively and moved once confirmed. A minimum dwell of 15 s (6 s for the next part of a full slide) prevents rapid changes, near-duplicates (similarity ≥ 0.85) are merged, and teacher edits are never overwritten. All thresholds were set by hand on D1 and are tested by ablation in Section IV.

### F. Composer and Image Retrieval

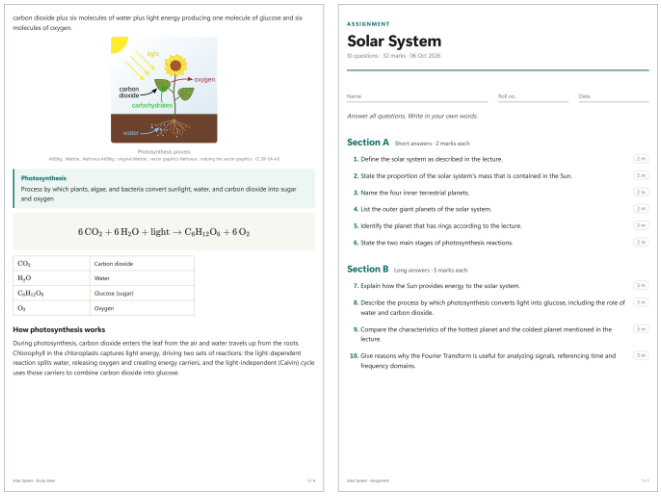
The composer maps items to definition cards, key points, fact tiles, process flows, comparison tables, timelines, cause-effect chains, classification trees and formulas; overflow moves to a further part (I, II, …) instead of shrinking text, and spoken mathematics is typeset with KaTeX. For a concrete object the LLM suggests a query; an openly licensed Wikimedia image (source, author and licence kept) is placed only if CLIP's zero-shot similarity [16] ranks it above generic distractor prompts ("a logo", "a map") and the query matches the slide text.

### G. Live Display and Teacher Control

A FastAPI server pushes versioned slide patches over WebSocket to /display (projector), /control (teacher) and /view (students, shared link). The control view (Fig. 2) adds editing, image search and upload, pause, blank screen, mistake cards, the running transcript and the lecture structure tree.

### H. Lecture Materials

After the lecture the event log is reused: on request it becomes study notes, an assignment (Fig. 3), a PPTX deck, or a live quiz or test answered through the shared link, all from the taught content and edited by the teacher. These features are not evaluated here.




(a) Study notes (b) Assignment

Fig. 3. Materials generated from the event log of a recorded D1 lecture (not evaluated): (a) a study-notes page; (b) the assignment.

## IV. Experimental Evaluation

### A. Setup and Data

Every run used one laptop (NVIDIA RTX 3050, 4 GB video memory; 16 GB system memory). D1 is six lectures spoken by the authors (114.1 min). The held-out set D2 (Tables II and III) has 30 lectures spoken live, unscripted, by six speakers into the laptop microphone (Recorded, physics to economics) and six 10-min public lectures played in as audio in real time (Public): MIT OpenCourseWare 7.016, 5.111, 18.01 and 8.01 (CC BY-NC-SA) and NPTEL Computer Networks and History of Economic Theory (CC BY-SA). The speakers consented to the recording. Against MIT's human captions of the 7.016 lecture (1,257 words), the live transcript has a word error rate of 7.1 %. Teachers did not edit slide text during a test; in D2 they deleted two items and accepted corrections from concern cards in two lectures. Three populations are evaluated: all 36 lectures for latency and operation, an annotated subset of 18 (12 Recorded, 6 Public; 191 min) for accuracy and the ablation, and 30 sampled slides for teacher ratings.

TABLE II
Evaluation Data

| Set | Lectures | Minutes | Units | Fallback units | Slides |
|---|---|---|---|---|---|
| D1 (development) | 6 | 114.1 | 389 | 45 | 163 |
| D2 Recorded | 30 | 349.2 | 1,275 | 59 | 453 |
| D2 Public | 6 | 60.0 | 344 | 13 | 52 |
| **D2 all (held out)** | **36** | **409.2** | **1,619** | **72** | **505** |

Units: groups of finished sentences sent for interpretation; fallback units were interpreted without the LLM (quota spent or provider failure).

TABLE III
Live System Performance (Session Logs)

| Measure (s) | D1 | D2 Rec. | D2 Pub. | D2 all |
|---|---|---|---|---|
| Speech-to-text per utterance | 0.66 | 0.71 | 0.60 | **0.68** |
| LLM call | 1.35 | 1.92 | 2.23 | **1.96** |
| Speech to slide, median | 12.5 | 10.1 | 17.1 | **11.2** |
| Speech to slide, 95th percentile | 36.8 | 23.2 | 93.4 | **53.0** |

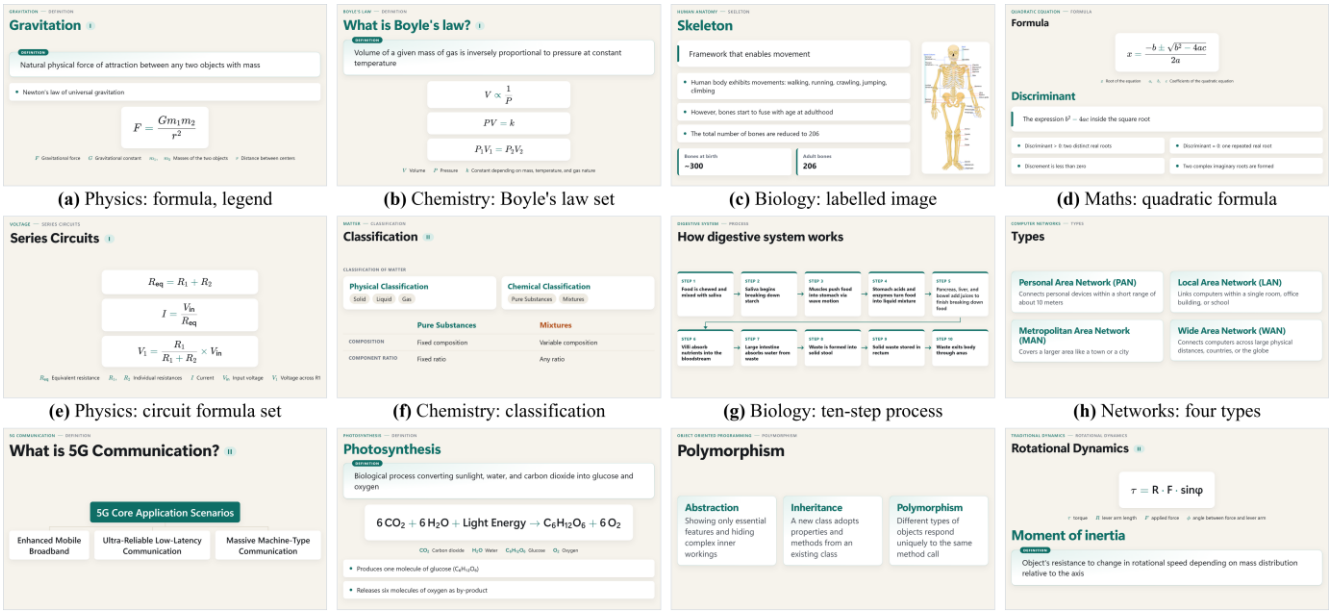
Medians unless stated; speech to slide: from the end of a spoken line to its display. Public includes web4 and web6 (provider outages).

*Annotation.* Two authors (A and B) independently annotated the 18 lectures, chosen to mix subjects and speakers with 6–17 min each. Before seeing the slides, each marked topic boundaries (where a teacher would start a new board) and key facts in the transcript, then judged whether each key fact was on a slide and whether each slide statement was said and correct; scores are the mean of A and B, without adjudication. A system boundary is the utterance that opened a new content slide. Boundaries are scored by precision, recall and F1 within ±1 utterance, averaged per lecture, and by $P_k$ [24], the probability that two utterances half a segment apart are wrongly split or joined; agreement is Cohen's κ [29] and, for skewed labels, Gwet's AC1 [30]. Intervals are 95 % bootstrap intervals over lectures. *Ratings.* Five teachers (T1–T5), not involved in development, rated 30 random D2 slides, each shown with what was said, from 1 to 5 on four criteria; their consistency is the ICC(2,1) of Shrout and Fleiss [31].

### B. Topic Boundaries and Content Accuracy

Slides change where the topic changes (Table IV): boundary precision is 0.78 (95 % interval 0.68–0.88), so a projected slide is rarely replaced within a topic, and $P_k$ is 0.32. Recall is 0.42 (0.30–0.55): a new concept is not a new slide, so the planner merges into the live slide sub-points that annotators marked as boundaries, most often in the Public lectures (recall 0.19); Section IV-C shows this is a trade-off. Of 669 key facts, 86.6 % (84.6–88.5 %) reached a slide (partly shown counts ½), and only 0.9 % (0.2–1.8 %) of 2,504 statement judgments were not supported by the speech. The coverage guard (3) is needed for this: 60 of the covered facts reached a slide only through a sentence it restored, so without it coverage would be 77.7 %; the guard adds 9.0 points (6.4–11.7). Of the 3.7 % marked wrong, 41 of 79 were also marked as said (a slip or mis-hearing): slides follow the speech, and corrections go to the teacher's concern cards; transcript support is therefore not factual correctness. Agreement is good for boundaries (κ = 0.72) and high for "supported" (88.8 %, AC1 0.88); κ is only 0.07 there because nearly all statements were supported, which lowers κ but not AC1 [30].

Test29, interpreted entirely by the fallback (quota spent), covered 83.1 % of key facts with 0.7 % unsupported statements but rarely opened a slide (recall 0.04) and used three layouts, 11 of its 13 slides plain points, against 12 layouts and 34 % plain points in the other lectures: the fallback preserves content, the LLM adds the structure. The 17 LLM lectures reach precision 0.77 and recall 0.45. The teachers rated the slides 3.87–4.17 of 5 (ICC(2,1) 0.45–0.56, moderate). In Fig. 4 the layout follows the kind of content, not the subject.




**(a)** Physics: formula, legend **(b)** Chemistry: Boyle's law set **(c)** Biology: labelled image **(d)** Maths: quadratic formula

**(e)** Physics: circuit formula set **(f)** Chemistry: classification **(g)** Biology: ten-step process **(h)** Networks: four types

**(i)** 5G: hierarchy tree **(j)** Biology: reaction, legend **(k)** OOP: three concepts **(l)** Physics: torque, definitions

Fig. 4. Projected D1 slides; (c), (d), (f) are app screenshots, the others re-render recorded LLM answers. Text is as spoken (the slip "Discrement" in (d)).

Table IV groups the results by what they measure. The Recorded lectures reach boundary precision 0.81, recall 0.54 and 88.0 % key-fact coverage; the Public lectures, university lectures recorded for online courses, reach 0.73, 0.19 and 84.0 %. In both groups fewer than 1 % of the slide statements were not said, and every mean rating is between 3.76 and 4.26.

TABLE IV
ACCURACY AND TEACHER RATINGS (18 ANNOTATED D2 LECTURES)

| Measure | Recorded | Public | All |
|---|---|---|---|
| *Topic boundaries* | | | |
| Precision ↑ | 0.81 | 0.73 | **0.78** |
| Recall ↑ | 0.54 | 0.19 | **0.42** |
| F1 ↑ | 0.59 | 0.29 | **0.49** |
| $P_k$ ↓ | 0.29 | 0.37 | **0.32** |
| *Slide content* | | | |
| Key facts on a slide ↑ | 88.0 % | 84.0 % | **86.6 %** |
| without coverage guard | 81.9 % | 69.1 % | 77.7 % |
| Statements not said ↓ | 0.9 % | 0.9 % | **0.9 %** |
| Statements marked wrong ↓ | 3.6 % | 4.0 % | **3.7 %** |
| *Teacher ratings (1–5)* ↑ | | | |
| Correctness | 4.20 | 4.06 | **4.15** |
| Relevance | 3.94 | 3.80 | **3.89** |
| Readability | 4.12 | 4.26 | **4.17** |
| Usefulness | 3.93 | 3.76 | **3.87** |
| *Annotator agreement* | | | |
| Boundaries, κ (raw) | 0.81 (94 %) | 0.56 (90 %) | 0.72 (92 %) |
| Statement said, AC1 (raw) | 0.89 (90 %) | 0.84 (85 %) | 0.88 (89 %) |

↑ / ↓: higher / lower is better. Means over annotators A and B; boundaries match within ±1 utterance; ratings: mean of T1–T5 over 30 slides (20 Recorded, 10 Public).

## C. Ablation of the Planner Rules

Table V replays the recorded signals and LLM answers of the 18 lectures through the planner with one rule changed (no new LLM calls; the unchanged replay reproduces the live decks). The continuity rule carries the result: if each concept opens a slide, the deck changes 3.2 times per minute, 200 slides are replaced within 10 s, and precision and $P_k$ drop to 0.37 and 0.47. The dwell time delays a slide change rather than cancelling it, so changes per minute stay at 1.15, but slides replaced within 10 s fall from 25 to 16 without loss of accuracy; a dwell of up to 20 s leaves precision and recall unchanged, while 45 s lowers them to 0.67 and 0.32. The topic confirmation brought no measurable benefit on D2 (precision 0.79 without it); it remains as a safeguard. With the LLM's labels deciding directly but slides still only extended, boundaries are similar: stability comes from the extend-only operations.

TABLE V
PLANNER ABLATION BY REPLAY (18 ANNOTATED D2 LECTURES)

| Planner variant | Precision ↑ | Recall ↑ | $P_k$ ↓ | Changes per min | Replaced < 10 s ↓ |
|---|---|---|---|---|---|
| **Full planner** (= live) | **0.78** | 0.42 | **0.32** | **1.15** | 16 |
| No dwell time | 0.78 | 0.42 | 0.32 | 1.15 | 25 |
| No topic confirmation | 0.79 | 0.45 | 0.31 | 1.12 | 10 |
| LLM labels decide | 0.78 | 0.46 | 0.30 | 1.17 | 21 |
| Each concept a slide | 0.37 | 0.86 | 0.47 | 3.18 | 200 |

Boundary scores as in Table IV; replaced < 10 s: slides replaced within 10 s of being shown.

## D. Comparison With an LLM-Controlled Baseline

TABLE VI
PROPOSED SYSTEM VS. LLM-CONTROLLED BASELINE (SAME D2 UNITS)

| Measure | Proposed | LLM baseline |
|---|---|---|
| Seen statements rewritten/removed ↓ | **2.0 %** | 38.2 % |
| Seen statements lost from final deck ↓ | **6.2 %** | 65.8 % |
| Lectures with fewer rewrites ↑ | **18 of 18** | 0 of 18 |
| Boundary precision ↑ | **0.78** | 0.72 |
| $P_k$ ↓ | **0.32** | 0.36 |
| Content slides per minute | 1.25 | 0.79 |

↑ / ↓: higher / lower is better. Rewritten: share of the statements of each replaced slide version that the next version changed or removed; lost: share of distinct shown statements absent from the final deck. Content slides per minute is not the changes per minute of Table V.

An LLM-controlled baseline without extend-only operations receives, for the same 850 units of the 18 annotated lectures, the slide it showed last and the last 60 s of transcript, and returns the slide to show, without planner, guards or composer (same two models, 0.5 M tokens). It placed slide changes about as well (Table VI; precision 0.72 vs. 0.78, $p = 0.34$) but rewrote or removed 1,439 of the 3,763 statements students had already seen (38.2 %), against 91 of 4,515 (2.0 %), and 65.8 % of what it showed was gone from its final deck, against 6.2 %. The difference holds in all 18 lectures: the mean per-lecture gap in rewrites is 39 points (95 % interval 32–47; exact paired sign-flip permutation test, $p < 0.001$). Of the proposed system's 91 changes, 24 extended a statement, 6 followed a correction the teacher accepted and 61 were composer merges. The comparison is between whole systems (the baseline also lacks the guards and composer); Table V isolates the planner rules.

### *E. End-to-End Latency*

Over all 36 D2 lectures (Table III), planning and composing take 2 ms. From the end of a spoken line to its content on the display the median is 11.2 s, of which 7.6 s is the deliberate wait in the gate for a complete idea; the system is live, not hard real-time. In web4 and web6 the free provider failed (timeouts, HTTP 429), units queued, and the 95th percentile rose to 53.0 s; without these two lectures the median is 10.2 s and the 95th percentile 23.4 s.

## V. Discussion and Conclusion

The Multimodal AI Teaching Copilot builds a continuity-aware deck from live speech with the LLM as a bounded interpreter; the planner's extend-only operations keep what students read stable (2.0 % vs. 38.2 % rewritten). Limitations: the annotators were authors (marking boundaries and facts before seeing the slides); recall is moderate by design; an append-only baseline prompt was not tested; the word error rate comes from the one lecture with human captions; the free provider caused outages; and without a classroom study no claim is made about learning. Anonymized annotations, ratings and evaluation scripts are available from the authors on request.

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